

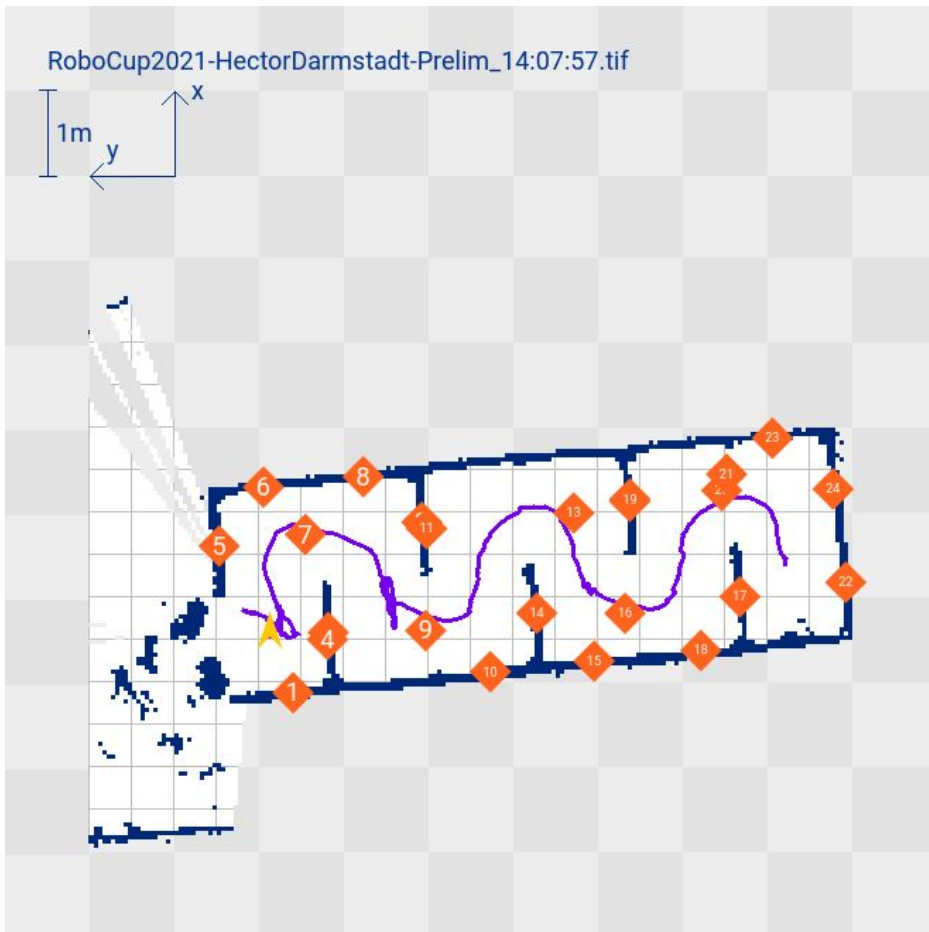
Mapping and Object Detection

Format Specifications

2D Map Format

2D maps have to be submitted in the specified [GeoTIFF](http://wiki.ros.org/hector_geotiff) format. An open-source implementation for ROS is available at http://wiki.ros.org/hector_geotiff.

Example map:



In the following, all elements of the map are described:

- **Fileformat:** [GeoTIFF](#)
- **FILENAME:** DARK BLUE (RGB: 0, 44, 207) TEXT
For example, "RoboCup2025-TeamName-Mission1.tiff" displayed in the upper left corner to identify the map, make it sort properly in a directory, and findable on a computer.
- **MAP SCALE:** DARK BLUE (RGB: 0, 50, 140) TEXT AND EXACTLY 1 METER LONG LINE
Display this in the upper right corner to indicate the scale of the map.
- **MAP ORIENTATION:** DARK BLUE (RGB: 0, 50, 140) TEXT ("X" AND "Y") AND ABOUT 50 cm LONG ARROWS
Display this next to the map scale. It gives the orientation for the victim location in the victim file. Must be a right handed coordinate system: X points upwards, Y to the left.
- **UNEXPLORED AREA GRID:** LIGHT/DARK GREY (RGB: 226, 226, 227/RGB: 237, 237, 238) CHECKERBOARD WITH 100CM SQUARES
This solid checkerboard pattern should show the unexplored area and provide scale on all sides of the mapped area. It should also print in black and white without ambiguity with other areas potentially turned grey in the process.
- **EXPLORED AREA GRID:** BLACK (RGB: 190,190,191) GRID WITH 50CM GRID AND ABOUT 1 CM THICK LINES (use a one pixel line in the map)
This grid should only appear in the explored area, behind any walls, victim locations, or other information. The grid should be aligned with the checkerboard pattern of the unexplored area, but twice as fine to allow visual inspection of wall alignments.
- **INITIAL ROBOT POSITION:** GREEN (RGB: 0, 240, 0) ARROW
This should mark the initial position of the robot and always be pointed toward the top of the map.
- **WALLS AND OBSTACLES:** DARK BLUE (RGB: 0, 40, 120) FEATURES
This should indicate the walls and other obstacles in the environment. The color should make the walls stand out from everything else.
- **SEARCHED AREA:** WHITE CONFIDENCE GRADIENT (RGB: 128, 128, 128 to RGB: 255, 255, 255)
This should be based on the confidence that the area is really free. It should produce a clean white when seen as free by all measurements and nearly untouched when undecided, that is, nearly equally seen as occupied as free, to produce a dither effect.
- **APRILTAG LOCATION:** SOLID YELLOW (RGB: 255, 200, 0) CIRCLE WITH ABOUT 35CM DIAM CONTAINING WHITE (RGB) TEXT "#" REPRESENTING THE LABEL ASSOCIATED WITH THE APRILTAG
- **HAZMAT LOCATION:** SOLID ORANGE (RGB: 255, 100, 30) DIAMOND WITH ABOUT 30CM SIDES CONTAINING WHITE (RGB) TEXT "@@" REPRESENTING THE FIRST TWO LETTERS OF THE HAZMAT LABEL
- **REAL OBJECT LOCATION:** SOLID RED (RGB: 240, 10, 10) DIAMOND WITH ABOUT 30CM SIDES CONTAINING WHITE (RGB) TEXT "@@" REPRESENTING THE FIRST TWO LETTERS OF THE IDENTIFIED OBJECT
- **ROBOT PATH:** MAGENTA (RGB: 120, 0, 140) LINE ABOUT 2CM THICK
This should show the robot path.

NOTE: If a team wishes to deviate from this 2D format in order to make **improvements** on the understandability and usefulness of their generated 2D map, they will not be penalized for doing so. The submitted map should contain all of the same information (scale, detections, physical barriers, robot path, etc.), but stylistic choices may be made to make it more presentable. The important capability to demonstrate is producing a clean, useful 2D map for field operations.

Identified Objects CSV Format

In addition to indicating identified objects in the map, they have to be submitted in a CSV list. In the following, the structure of the file is given:

Filename: RoboCup[Year]-[Teamname]-[Mission]-[Start Time]-pois.csv

([Start Time] should be formatted as HH-MM-SS)

Format for the file header:

```
"pois"
"1.3"
"[Your team name]"
"[Your country]"
"[Start Date]"
"[Start Time]"
"[Mission #]"

detection,time,type,name,x,y,z,robot,mode
```

Replace the placeholders "[...]" with your data. Please leave the quotes in the file, but not the brackets.

The file body contains one line for each found object in the following format:

```
[detection],[time],[type],[name],[x],[y],[z],[robot],[mode]
```

- **[detection]:** Unique integer counter for each object that is also printed on the geotiff map
- **[time]:** Time stamp when the object was found
- **[type]:** Type of the object: [ar_code, hazmat_sign, real_object, heat_sig]
- **[name]:** Unique ID for detection (i.e. ar_code number, hazmat label, object name)
- **[x], [y], [z]:** Coordinates of the object in meters
- **[robot]:** Name of the robot that found the object
- **[mode]:** Mode of the robot when it found the object: **A** for autonomous, **T** for teleoperated

Strings which include spaces should be enclosed with double quotes. Here is an example of a file which might be named "RoboCup2025-ReskoKoblenz-Prelim1-pois.csv":

```
"pois"
"1.3"
"Resko Koblenz"
"Germany"
"2018-06-23"
"14:37:03"
"Prelim1"

detection,time,type,name,x,y,z,robot,mode
1,14:28:01,"ar_code","2",-8.2992,-2.2904,0.49,"Robbie 1",A
2,14:28:02,"ar_code","34",-8.2993,-2.2902,0.4563,"Robbie 2",T
3,14:28:05,"hazmat_sign","poison",-5.7452,-7.0849,0.2130,"Robbie 1",A
4,14:30:14,"real_object","gloves",-6.0845,-0.1251,0.3561,"Robbie 1",A
5,14:32:56,"heat_sig","0",-9.2817,0.04968,0.7561,"Robbie 2",T
```

IMPORTANT: There will only be **one** instance of each object included in the mapping area at any time. Teams must only include **one** detection per unique object. If duplicate objects are contained in the submitted CSV, only the first instance of that object will be used for scoring.

A list of object labels will be provided to teams. **Mislabelled objects and submissions not following the specified format will not be scored.**

3D Point Cloud Map

Teams will submit a 3D pointcloud representation of the mapped area to be scored with respect to a generated ground truth map. The submitted map should adhere to the following specification:

Filename: RoboCup[Year]-[Teamname]-[Mission]-[Start Time]-map.ply
([Start Time] should be formatted as HH-MM-SS)

Your point cloud should be formatted as an ASCII file.

The header of your file should adhere to the PLY file format.

At a minimum, your file should include float type fields for x, y, and z.

Teams may include additional fields such as:

- RGB Color Information
- Heat Information
- Point Normal Information
- Confidence Scalar

Maps which contain sensible color and/or heat data will be awarded a bonus point multiplier.

IMPORTANT: The scale of your map coordinates are expected to be in meters. (i.e. A distance of 1 unit in the submitted point cloud should correspond to 1 meter in the real world). **Improper scaling will severely penalize your score.**

The origin location (0,0,0) of the 3D Cloud should be set to the starting position for the robot which will be marked on the floor in the mapping area. More specifically, the origin should be set to be the center of the front of the robot, at floor height. Rotationally, the starting position will point the robot in the +Y direction, and the vertical axis will be Z.

More information on PLY format can be found [here](#).

The format for the PLY header that should be included is shown below.

NOTE: You are not required to include all fields if you are not collecting those kinds of data

```
ply
format ascii 1.0
comment {Team Name}
comment {Start time}
comment {Mission #}
element vertex {Number of Vertices}
property float x
property float y
property float z
property uchar red
property uchar green
property uchar blue
property float nx
property float ny
property float nz
property float temp
property float confidence
end_header
{DATA}
```